Summary in Graph

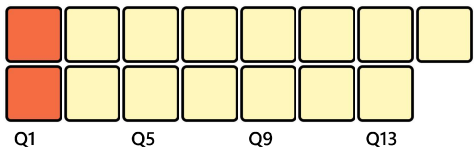
# Exam Summary (GO Classes CS/DA Test Series 2026 | Databases | Topic Wise Test)

| | | | |
|---|---|---|---|
| **Qs. Attempted:** | **4**<br>4 + 0 | **Correct Marks:** | **0**<br>0 + 0 |
| **Correct Attempts:** | **0**<br>0 + 0 | **Penalty Marks:** | **0.67**<br>0.67 + 0 |
| **Incorrect Attempts:** | **4**<br>4 + 0 | **Resultant Marks:** | **-0.67**<br>-0.67 + 0 |

| | |
|---|---|
| **Total Questions:** | **15**<br>5 + 10 |
| **Total Marks:** | **25**<br>5 + 20 |
| **Exam Duration:** | **45 Minutes** |
| **Time Taken:** | **45 Minutes** |

## Individual Question Statistics

Select Metric: Marks Obtained




* Average Stats based on first-attempt stats of Top 10% (244) out of 2443 users

EXAM RESPONSE | EXAM STATS | FEEDBACK

## Technical

Q #1 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Databases

What does physical data independence allow the database administrator to do?

A. Change the conceptual schema without changing the internal schema
B. Change the internal schema without changing the conceptual schema
C. Change the external schema without changing the conceptual schema
D. Change the application programs without changing the internal schema

Your Answer: C | Correct Answer: B | Incorrect | Time taken: 00min 44sec | Discuss

Q #2 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Databases

In a relation, $\mathrm{R}$, an attribute "B" of $\mathrm{R}$ is "fully functional dependent" on an attribute or set of attributes "A" of $\mathrm{R}$ if $\mathrm{B}$ is functional dependent on $\mathrm{A}$ but not functional dependent on any proper subset of $\mathrm{A}$. Consider a relation $\mathrm{S}$ in which every non-prime attribute is "fully functional dependent" on every prime attribute. What is necessarily true for $\mathrm{S}$?

A. $\mathrm{S}$ is in $2\mathrm{NF}$.
B. $\mathrm{S}$ is not in $2\mathrm{NF}$.
C. $\mathrm{S}$ is in $2\mathrm{NF}$ but not in $3\mathrm{NF}$.
D. None of the above.

Your Answer: A | Correct Answer: D | Incorrect | Time taken: 03min 39sec | Discuss

Q #3 | Multiple Select Type | Award: 1 | Penalty: 0 | Databases

Which of the following is/are NOT a valid rule about functional dependencies (FD's) in context of RDBMS model?

A. If $\mathrm{A} \rightarrow \mathrm{B}$ then $\mathrm{B} \rightarrow \mathrm{A}$
B. If $\mathrm{AB} \rightarrow \mathrm{C}$ and $\mathrm{A} \rightarrow \mathrm{C}$, then $\mathrm{B} \rightarrow \mathrm{C}$
C. If $\mathrm{AB} \rightarrow \mathrm{C}$, then $\mathrm{A} \rightarrow \mathrm{C}$ or $\mathrm{B} \rightarrow \mathrm{C}$
D. If $\mathrm{A} \rightarrow \mathrm{BC}$, then $\mathrm{A} \rightarrow \mathrm{C}$ and $\mathrm{A} \rightarrow \mathrm{B}$

Your Answer: A;D | Correct Answer: A;B;C | Incorrect | Time taken: 03min 43sec | Discuss

Q #4 | Numerical Type | Award: 1 | Penalty: 0 | Databases

Consider a relation scheme $\mathrm{R}(\mathrm{A}, \mathrm{B}, \mathrm{C}, \mathrm{D}, \mathrm{E})$ with three candidate keys $\{\mathrm{AB}, \mathrm{BC}, \mathrm{CD}\}$. How many super keys are possible?

Your Answer: 14 | Correct Answer: 16 | Incorrect | Time taken: 06min 19sec | Discuss

Q #5 | Multiple Select Type | Award: 1 | Penalty: 0 | Databases

Given a relation schema $\mathrm{R}$ with a set $\mathrm{F}$ of $\mathrm{FD}$s that hold over $\mathrm{R}$, let's define $\mathrm{X}$ to be a subset of the attributes of $\mathrm{R}$, and $\mathrm{A}$ is an attribute of $\mathrm{R}$. If, for every $\mathrm{FD}$ $\mathrm{X} \rightarrow \mathrm{A}$ in $\mathrm{F}$, one of the following is true: $\mathrm{A} \in \mathrm{X}$ (reads $\mathrm{A}$ is an element of $\mathrm{X}$) - i.e. it is a trivial $\mathrm{FD}$ or $\mathrm{X}$ is a superkey or $\mathrm{A}$ is a part of some key for $\mathrm{R}$ Then $\mathrm{R}$ is in which form?

A. $1\mathrm{NF}$
B. $2\mathrm{NF}$
C. $3\mathrm{NF}$
D. $\mathrm{BCNF}$

Your Answer: | Correct Answer: A;B;C | Not Attempted | Time taken: 30min 35sec | Discuss

Q #6 | Numerical Type | Award: 2 | Penalty: 0 | Databases

Consider a relation $\mathrm{R}(\mathrm{A}, \mathrm{B}, \mathrm{C}, \mathrm{D}, \mathrm{E})$ with functional dependencies:

$$\mathrm{A}, \mathrm{B} \rightarrow \mathrm{C}$$
$$\mathrm{C}, \mathrm{D} \rightarrow \mathrm{E}$$

Suppose there are at most 3 different values for each of $\mathrm{A}$, $\mathrm{B}$, and $\mathrm{D}$. What is the maximum number of different values for $\mathrm{E}$?

Your Answer: | Correct Answer: 27 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Numerical Type | Award: 2 | Penalty: 0 | Databases

Consider a relation schema $\mathrm{R}$ with attributes $(\mathrm{A}, \mathrm{B}, \mathrm{C}, \mathrm{D}, \mathrm{E}, \mathrm{F})$ and the functional dependency set $\mathrm{F}$ that holds on $\mathrm{R}$ :

$$\mathrm{F} = \{\mathrm{F} \rightarrow \mathrm{AE}, \mathrm{A} \rightarrow \mathrm{B}, \mathrm{B} \rightarrow \mathrm{ADE}, \mathrm{AD} \rightarrow \mathrm{B}, \mathrm{BDE} \rightarrow \mathrm{AD}\}.$$

How many super keys $\mathrm{R}$ has?

Your Answer: | Correct Answer: 16 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Numerical Type | Award: 2 | Penalty: 0 | Databases

A database relation $\mathrm{R}$ has 10 attributes $\{\mathrm{A}, \mathrm{B}, \mathrm{C}, \mathrm{D}, \ldots, \mathrm{J}\}$. "CDEF" is known to be a candidate key of relation $\mathrm{R}$. There may be other candidate keys, which are unknown. What is the maximum number of candidate keys that $\mathrm{R}$ could simultaneously have?

Your Answer: | Correct Answer: 247 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Select Type | Award: 2 | Penalty: 0 | Databases

Consider a relation schema $\mathrm{R}$ with attributes $(\mathrm{X}, \mathrm{Y}, \mathrm{Z}, \mathrm{U}, \mathrm{V}, \mathrm{W})$ and the functional dependency set $\mathrm{F}$ that holds on $\mathrm{R}$ :

$$\mathrm{F} = \{\mathrm{Y} \rightarrow \mathrm{U}, \mathrm{V} \rightarrow \mathrm{W}, \mathrm{U} \rightarrow \mathrm{V}, \mathrm{U} \rightarrow \mathrm{Y}, \mathrm{W} \rightarrow \mathrm{YU}\}.$$

$\mathrm{R}$ is decomposed into two relations $\mathrm{P}(\mathrm{X}, \mathrm{Y}, \mathrm{U}, \mathrm{V}), \mathrm{S}(\mathrm{Y}, \mathrm{Z}, \mathrm{U}, \mathrm{W})$.
Which of the following is/are true regarding this decomposition?

A. Decomposition is lossless.
B. Decomposition is dependency preserving.
C. $\mathrm{P}$ is in $3\mathrm{NF}$.
D. $\mathrm{S}$ is in $3\mathrm{NF}$.

Your Answer: | Correct Answer: B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Select Type | Award: 2 | Penalty: 0 | Databases

Given the relation schema $\mathrm{R} = (\mathrm{A}, \mathrm{B}, \mathrm{C}, \mathrm{D}, \mathrm{E})$ and the set of functional dependencies (FDs):

$$\mathrm{F} = \{\mathrm{E} \rightarrow \mathrm{AB}, \mathrm{BC} \rightarrow \mathrm{D}, \mathrm{D} \rightarrow \mathrm{E}, \mathrm{AB} \rightarrow \mathrm{BC}, \mathrm{BC} \rightarrow \mathrm{E}\}$$

For which of the following FD "S", closure of F is same as closure of F-S? i.e. which of the following FD can be removed from F without affecting the Closure of the Set of Functional Dependencies F?

A. $AB \rightarrow B$
B. $AB \rightarrow C$
C. $BC \rightarrow E$
D. $BC \rightarrow D$

Your Answer: | Correct Answer: A;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 | Multiple Select Type | Award: 2 | Penalty: 0 | Databases

Consider the relational schema $(A, B, C, D)$ with Functional Dependencies set $F = \{A \rightarrow B, B \rightarrow ACD, CD \rightarrow B\}$;

Which of the following Functional Dependencies set is equivalent to F?

A. $\{A \rightarrow B, B \rightarrow CD, CD \rightarrow A\}$
B. $\{A \rightarrow CD, B \rightarrow A, CD \rightarrow B\}$
C. $\{A \rightarrow BCD, B \rightarrow A, CD \rightarrow A\}$
D. $\{A \rightarrow CD, B \rightarrow CD, CD \rightarrow AB\}$

Your Answer: | Correct Answer: A;B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Databases

Consider the relational table $R(P, Q, S, T, U, V, W, Y)$ and the following functional dependencies set F over R :

$$P \rightarrow QST, PT \rightarrow U, UVW \rightarrow Y, V \rightarrow WY$$

Which of the functional dependencies can be removed from this FD set F without altering the candidate keys (i.e. after removal of the FD, set of candidate keys in resulting relation must be the same as set of candidate keys in R)?

A. $P \rightarrow QST$
B. $PT \rightarrow U$
C. $UVW \rightarrow Y$
D. $V \rightarrow WY$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Databases

For a relation R, let $\alpha$ and $\beta$ be sets of attributes such that $\alpha \rightarrow \beta$ holds, but $\beta \rightarrow \alpha$ does not hold. Let A be an attribute that is not in $\alpha$, is not in $\beta$, and for which $\beta \rightarrow A$ holds. Then we say that A is "mysteriously dependent" on $\alpha$.

A relation schema R is "_____" with respect to a set F of functional dependencies if there are no non-prime attributes A in R for which A is "mysteriously dependent" on some key for R.

What is the correct term to put in the "_____"?

A. 3NF

B. Not $3NF$
C. $BCNF$
D. Not $2NF$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Databases

Consider the relation $R(A, B, C, D, E)$ with the following functional dependencies:

$$A \rightarrow C; B \rightarrow C; C \rightarrow D; DE \rightarrow C; CE \rightarrow A$$

Consider the decomposition "DECOMP" of the relation $R$ into five relations :

$$DECOMP = \{R1, R2, R3, R4, R5\}$$

Where, $R1 = \{A, D\}; R2 = \{A, B\}; R3 = \{B, E\}; R4 = \{C, D, E\}; R5 = \{A, E\}$

Which one of the following options is correct?

A. DECOMP is a lossless decomposition, but not dependency preserving.
B. DECOMP is a lossless decomposition, also dependency preserving.
C. DECOMP is a lossy decomposition, but not dependency preserving.
D. DECOMP is a lossy decomposition, but dependency preserving.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Databases

Consider the following two statements:

1. Suppose we are given a relation $R$ with attributes $A, B, C, D$ and FDs :

$$B \rightarrow C; A \rightarrow B; A\ C \rightarrow D$$

Then this FD set is irreducible i.e. Minimal.

2. Suppose a relation schema $S(A, B, C, D, E, F, G, H, I, J)$ has FD set :

$$A\ B\ D \rightarrow E\ ;\ A\ B \rightarrow G\ ;\ B \rightarrow F\ ;\ C \rightarrow J\ ;\ C\ J \rightarrow I\ ;\ G \rightarrow H$$

Then this $FD$ set is irreducible/minimal.

Which of the above statements is/are false?

A. $1$ Only
B. $2$ Only
C. Both
D. None.

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Copyright & Stuff